Nagar Yuwak Shikshan Sanstha's
Yeshwantrao Chavan College of Engineering
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Department of Computer Technology

Vision of the Department

To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.

Mission of the Department

To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary area by developing Problem-solving skills through emerging technologies.

Session 2026-2027

Vision: Dream of where you want.	**Mission:** Means to achieve Vision

Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	**Preparation**	**P: Preparation**	**Pep-CL abbreviation pronounce as Pep-si-lL easy to recall**
PEO2	**Core Competence**	**E: Environment (Learning Environment)**	
PEO3	**Breadth**	**P: Professionalism**	
PEO4	**Professionalism**	**C: Core Competence**	
PEO5	**Learning Environment**	**L: Breadth (Learning in diverse areas)**	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:
Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and Finance, Life-Long Learning

PSO Keywords:

"I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life." *to contribute to the development of cutting-edge technologies and Research*.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

Krishna Mishra
Name and Signature of Student and Date
(Signature and Date in Handwritten)

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Session	**2026-27 (ODD)**	**Course Name**	**Lab: Generative AI**
Semester	**7**	**Course Code**	**23ADS1706**
Roll No	**41**	**Name of Student**	**Krishna Mishra**

Practical Number	**8**
Course Outcome	CO1: Explain the fundamentals of Artificial Intelligence, Neural Networks, Natural Language Processing, and Transfer Learning used in Generative AI applications. CO2: Apply Generative AI models such as GANs, VAEs, Transformers, BERT, GPT, and T5to solve real-world AI and NLP problems. CO3: Analyze and optimize prompts using prompt engineering techniques for improving the performance of Large Language Models in different application domains. CO4: Design and develop Generative AI solutions by integrating Foundation Models, Retrieval-Augmented Generation (RAG), tokenization, fine-tuning, and chunking techniques for domain-specific applications.
Aim	To develop a Generative AI chatbot that generates emails, reports and technical explanations using an LLM and prompt engineering techniques.
Problem Definition	Develop a Generative AI chatbot capable of generating emails, reports, and technical explanations using an LLM with prompt engineering techniques.
Theory (100 words)	A Large Language Model (LLM) generates text by predicting the next token based on the given prompt. Prompt engineering is the technique of writing clear prompts to get better outputs. Important techniques are role prompting ("You are a technical writer"), clear task instructions, output constraints (length, structure, tone) and the use of examples. In this experiment, the instruction-tuned Flan-T5 model is used with separate prompt templates for emails, reports and explanations. The user selects the task and topic, the template is filled, the model generates the content and the chatbot displays it.
Procedure	Algorithm: • Start.

and
Execution
(100 Words)

- Load the pre-trained Flan-T5 model and its tokenizer.
- Create prompt templates for email, report and explanation with a role, task and constraints.
- Take the task type and topic from the user.
- Fill the selected template with the user's inputs to form the prompt.
- Tokenize the prompt and generate the text using beam search.
- Decode and display the generated content.
- Repeat until the user types exit, then stop.

Code:

```
import torch
from transformers import AutoTokenizer,
AutoModelForSeq2SeqLM

model_name = "google/flan-t5-base"

tokenizer = AutoTokenizer.from_pretrained(model_name)
model =
AutoModelForSeq2SeqLM.from_pretrained(model_name)


templates = {
    "email": (
        "You are a professional office assistant. Write a formal "
        "and polite email to {to} about {topic}. Include a "
        "subject, greeting, body and closing. Keep it under 120
words."
    ),
    "report": (
        "You are a technical writer. Write a short report on {topic}
"
        "with an introduction, three key points and a conclusion."
    ),
    "explain": (
        "You are a patient teacher. Explain {topic} in simple "
        "words for a beginner engineering student and give one
example."
    )
}


def generate(prompt, max_new_tokens=250):
    inputs = tokenizer(prompt, return_tensors="pt")

    with torch.no_grad():
        outputs = model.generate(
            **inputs,
            max_new_tokens=max_new_tokens,
            num_beams=4,
            no_repeat_ngram_size=3,
            early_stopping=True
        )

    return tokenizer.decode(
        outputs[0],
```

```
        skip_special_tokens=True
    )


def chatbot():
    print("AI Content Generation Chatbot (type 'exit' to quit)")

    while True:
        task = input("\nChoose task [email/report/explain]:
").strip().lower()

        if task == "exit":
            print("Chatbot closed.")
            break

        if task not in templates:
            print("Please enter email, report or explain.")
            continue

        topic = input("Enter the topic: ").strip()

        if not topic:
            print("Topic cannot be empty.")
            continue

        if task == "email":
            recipient = input("Recipient: ").strip()

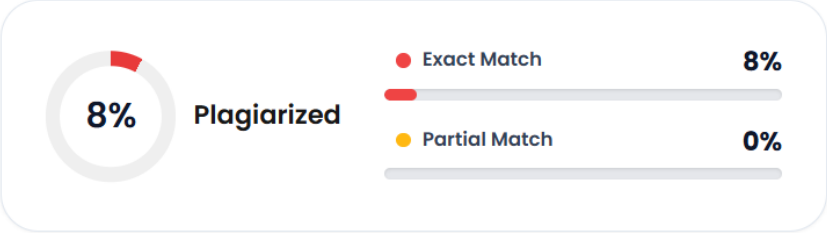
            if not recipient:
                print("Recipient cannot be empty.")
                continue

            prompt = templates[task].format(
                to=recipient,
                topic=topic
            )
        else:
            prompt = templates[task].format(topic=topic)

        print("\nGenerated Content:\n")
        print(generate(prompt))


# Demonstration with fixed inputs
demo = [
    templates["email"].format(
        to="the Head of Department",
        topic="a request for leave on Friday"
    ),
    templates["report"].format(
        topic="the benefits of cloud computing"
    ),
    templates["explain"].format(
        topic="overfitting in machine learning"
    )
]

for prompt in demo:
    print("PROMPT:", prompt)
```

	print("OUTPUT:", generate(prompt)) print() # Uncomment the next line to run the interactive chatbot # chatbot()
Output Analysis	• Email: For the prompt on a leave request to the Head of Department, the chatbot generates a polite email with a subject, greeting, short body and closing. • Report: For the topic "benefits of cloud computing", the chatbot generates a short report containing an introduction, key points and a conclusion. • Explanation: For the topic "overfitting in machine learning", the chatbot generates a simple explanation with an example. • The role, task and constraint parts of the prompt control the style and length of the generated text. Record the generated outputs from your run here.
Link of student Github profile where lab assignment has been uploaded. Plag Report (Similarity index < 12%)	Github.com/krish-na45/GenAI-Practicals 
Conclusion	The experiment developed a Generative AI chatbot that generates emails, reports and technical explanations using an LLM. Using prompt engineering techniques such as role, task instructions and constraints improved the relevance and structure of the generated content. The same prompt templates can also be used with larger LLMs to obtain longer and higher-quality outputs.
Date	/ /2026 .